data is continuously uploaded to the cloud, creating a digital image of the 'behaviour' of the city, available online at every moment. This digital shadow is possible because of the latest achievements in connectivity, especially the Internet of Things — a set of technologies that connects 'intelligent' objects through the Internet, adding another layer of data.

The resulting corpus of data becomes the foundation for many other services and products, created by institutions and companies that might build second-layer solutions. For instance, an app could be created that allows citizens to improve their mobility with eco-friendly and connected solutions. Another app could help building owners improve energy efficiency through data and algorithms that control heating and cooling in real-time. A service that connects government officials and citizens, improving their communication by digital means, could be offered by an app too. Other applications might be related to urban farming, tele-health, air quality, data management, or even toilets-as-a-service!

The range of potential solutions that can be built over the digital shadow is vast and is continually growing. These solutions can be broken down into the following six areas:

1. Smart environment - minimising the ecological footprint of a city.
2. Smart living - improving the quality of life in the city.
3. Smart economy - increasing the city's competitiveness.
4. Smart mobility - creating new and sustainable mobility options in the city.
5. Smart government and governance - promoting the participation of citizens and increasing transparency in the public sector.
6. Smart people - preserving and increasing the human capital of a city, helping people to develop their full potential and fostering their participation in social life.

Smart city management model (SCMM)

The SCMM, shown in Figure 1, addresses the main dimensions of any smart city project:

- The services areas, in the centre of the figure, summarise the kind of initiatives undertaken in a smart city project.
- The transformation cycle process, depicted as a circle in Figure 1, shows the necessary steps needed to succeed when building a smart city programme: initiate transformation, determine location, develop concepts and synchronise partners, activate resources, and realise projects as well as operation and instrumentation.
- The fundamental elements, shown around the perimeter of the figure, are data governance, partner and city participation, financing, technology, business models and communication.

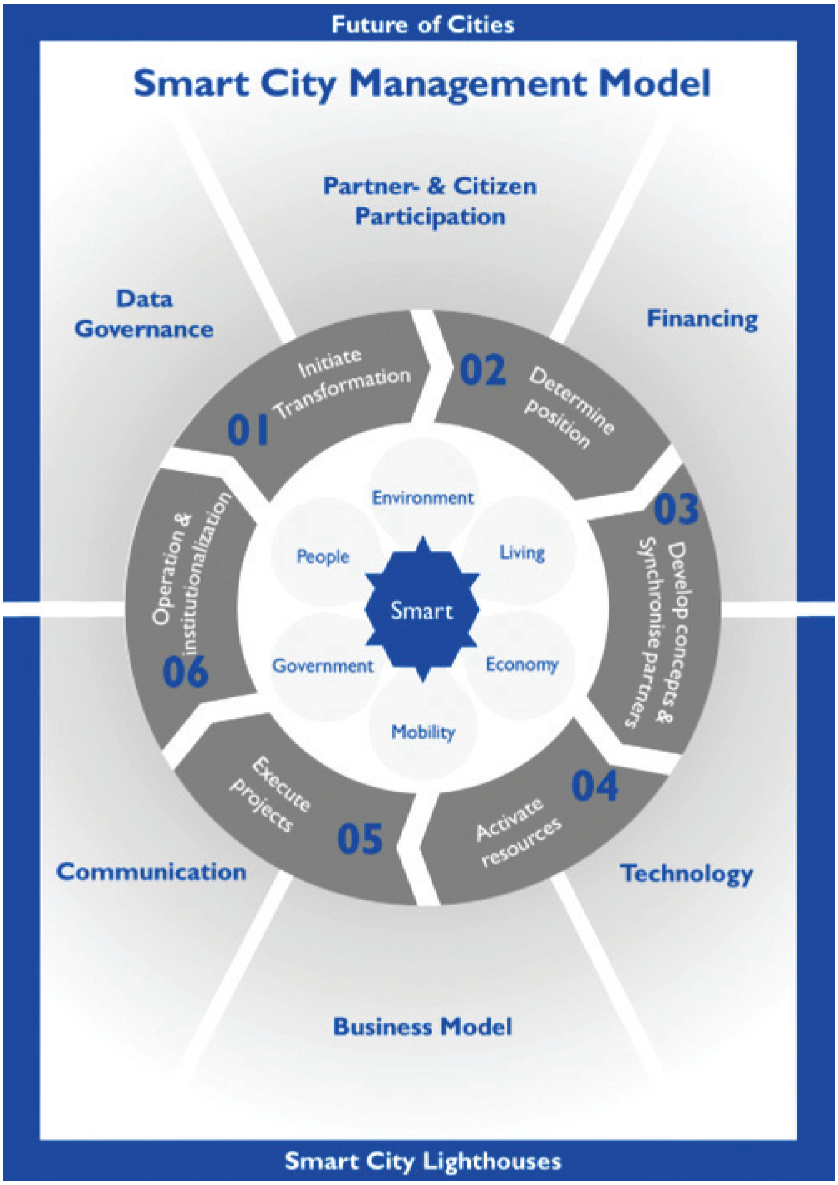

Figure 1: Smart city management model